

Mapping the potential public and private returns to Enhanced Rock Weathering in Sub-Saharan Africa

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Abstract

Enhanced Rock Weathering (ERW) – spreading finely ground silicate rock such as basalt on cropland – sits at an unusual intersection of two value streams: it durably removes atmospheric CO₂ (a public good, monetizable through voluntary carbon markets, VCM) and it neutralizes soil acidity much like agricultural lime, raising yields (a private return to the farmer). We develop a spatially explicit, decomposed *ex ante* profitability model for 44 Sub-Saharan African (SSA) countries and 23 crops that separates the per-hectare gross margin of basalt-ERW into an agronomic (private) term and a carbon (public) term, each net of the full delivered cost. We map three decision-relevant envelopes: land where the private return alone justifies investment, land where the public return alone justifies it, and their intersection. Crediting only durably exported carbon – deducting the alkalinity consumed neutralizing soil acidity, which re-releases its CO₂ just as lime does – we evaluate the public (carbon) and private (yield) cases together in the equilibrium steady state. Under a spatially targeted lime-equivalent

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dose the private envelope (4.3 Mha) exceeds the durable public envelope (1.7 Mha), and the two coincide on 1.2 Mha; but the *combined* envelope is far larger (6.9 Mha), because on roughly 2.1 Mha neither stream alone covers the cost yet together they do. This is the central economic result: voluntary carbon finance is not a bonus on an already-profitable practice but the margin that makes an otherwise-unprofitable agromonic investment viable. Properly accounting for acidity neutralization cuts credited removal by about 80% at first application and shows that durable removal is largely a steady-state phenomenon; the carbon case is bound by the realized CO₂ removed per tonne of rock (median marginal abatement cost \$446/t CO₂), the largest and least-resolved lever. We map a robust core-targeting geography and the field-data priorities that would most improve it.

Keywords: enhanced rock weathering; carbon dioxide removal; voluntary carbon markets; soil acidity; agricultural liming; Sub-Saharan Africa; additionality; *ex ante* analysis.

JEL: Q12, Q15, Q54, Q56, O13.

1 Introduction

Enhanced rock weathering (ERW) has attracted considerable attention as a carbon dioxide removal (CDR) pathway that promises climate mitigation while potentially delivering material benefits to farmers (Beerling et al., 2020, 2025b). The mechanism is well understood. Silicate rock – most commonly basalt – is quarried, ground to a fine dust, and spread on agricultural fields. There, rainwater and the CO₂ respired by roots and soil microbes form carbonic acid, which dissolves the silicate minerals; this reaction consumes CO₂ and converts it into dissolved bicarbonate (HCO₃[−]) while releasing base cations such as Ca²⁺ and Mg²⁺. The bicarbonate is carried by drainage water through rivers to the ocean, where the captured carbon is stored on geological timescales (in excess of 10,000 years) (Beerling et al., 2020; Renforth, 2019). The same reaction that sequesters carbon also neutralizes soil acidity: the

released cations raise soil pH and base saturation, relieving the aluminum and manganese toxicity and nutrient-availability constraints that depress yields on acid soils, so that ERW can raise crop yields much as agricultural lime does (Kantola et al., 2017; Skov et al., 2024).

While the mechanisms are well established, how ERW would play out in practice – particularly in data-limited geographies – is far less substantiated. In particular, the economics that underlie return-on-investment calculations have not been evaluated in a spatially explicit, granular way, which makes it difficult to know how and where to invest. This gap matters because the public investment case, tied to carbon removal, may be spatially distinct from the private case – the locations where the value of yield gains exceeds the cost of applying ERW. Where these two return surfaces diverge, alternative funding strategies – voluntary carbon market (VCM) financing, private input-market development, or some blend of the two – face very different spatial opportunity sets. The absence of explicit, comparable estimates is thus a barrier to investment and targeting. One concrete version of the targeting problem is the question a carbon-market investor would ask: where can payments for carbon sequestration also deliver benefits to smallholder farmers?

In this paper we take a spatially explicit, *ex ante* modeling approach to characterize the benefit streams that ERW would generate across different geographical contexts in Sub-Saharan Africa (SSA), an understudied region with large potential payoffs to the practice. Our approach is grounded in the observation that the economics of ERW are fundamentally governed by spatial conditioners: CDR potential depends on soil chemistry and rainfall; the yield dividend from acidity neutralization varies with crop, soil type, and baseline acidity; and the delivered cost of feedstock to the farm gate is built up from quarry proximity, grinding energy, and transport distance. Bringing these spatially varying factors into a single accounting framework, we evaluate how the distributions of public and private returns compare with one another and with the site-specific delivered cost of ERW.

Our work draws on and connects two largely separate literatures, reviewed in Section 2: a fast-growing body that substantiates the geochemistry and carbon-removal potential of

ERW but rarely examines its on-farm economics, and a spatially explicit agronomic and economic literature on the returns to soil-acidity remediation through liming. The few ERW yield trials in Sub-Saharan Africa report responses in relative terms over short horizons, without economic returns (Oppong Danso et al., 2025; Akortey et al., 2026), while the liming literature (Vasco Silva et al., 2025; Hijbeek et al., 2021) supplies the integrated, spatially explicit framework we adapt.

Our value-added is to take that liming framework and reconfigure it for ERW, accounting jointly for the agronomic yield benefit and the carbon-removal benefit, net of the spatially varying delivered cost of feedstock, while operationalizing the mechanistic CDR understandings from the geochemical literature at high spatial resolution. To our knowledge this is the first analysis to map the private and public benefit streams of ERW side by side and net of location-specific delivery costs. Doing so lets us delineate the opportunity space for carbon-market financing, which is especially attractive where it also delivers smallholder co-benefits, and ask where the private return alone is large enough that a market for ERW could be built on farmers’ own willingness to pay.

We address six questions, the first three scoping the financing opportunity and the last three turning the estimates into decisions and a model-improvement agenda.

1. Where is there a case for a *private market* for ERW, regardless of public returns – that is, where do yield-based returns alone exceed the delivered cost, so that farmers or input suppliers could fund deployment without carbon finance?
2. Where is there a case for *voluntary-carbon-market (VCM) investment*, regardless of private returns – where does the value of durable carbon removal alone exceed the delivered cost?
3. Where do the two *coincide*, strengthening the proposition for a carbon buyer – where a purchased tonne of removal also delivers a smallholder yield co-benefit?
4. How robust are these answers to parameter uncertainty (delivered cost, carbon price,

yield response, CDR rate, and the carbon-accounting choice)?

5. Given that uncertainty, which areas should early *pilots* target?

6. Which assumptions and processes – and in which areas – would targeted *field data* most improve, to drive an iterative modeling loop?

A point on framing: the private return and the carbon-market *co-benefit* are the same quantity – the economic value of ERW-induced yield gains – seen from two vantage points, the smallholder who captures it directly and the carbon buyer who values it as bundled impact. Questions 1–3 map onto the private, public, and intersection return surfaces we estimate.

The remainder of the paper is organized as follows. Section 2 reviews the relevant literature in more detail. Section 3 sets out our conceptual framework and modeling approach. Section 4 presents our core results and the sensitivity of our conclusions to parameter uncertainty. Section 5 discusses how the results may inform investment strategies, pilot-program targeting, and further research, and Section 6 offers concluding remarks.

2 Background and related literature

ERW as a joint product. Enhanced rock weathering is distinctive among carbon-removal options in producing two benefits from a single action. When crushed silicate rock, typically basalt, dissolves in soil, the reaction consumes carbonic acid and converts CO₂ into bicarbonate while releasing base cations such as Ca²⁺ and Mg²⁺ (Beerling et al., 2020; Kantola et al., 2017). The bicarbonate, once it leaches to rivers and the ocean, stores carbon on geological timescales; the same cations raise soil pH and relieve the aluminum and manganese toxicity that depress yields on acid soils, so the agronomic effect closely parallels agricultural lime (Lewis et al., 2021). The contrast with lime is instructive, since lime’s alkalinity carries carbon that is re-released when it neutralizes acidity, making liming carbon-neutral at best,

whereas the silicate’s alkalinity is generated by consuming atmospheric CO₂. Field trials increasingly document the yield side of this joint product, in temperate systems (Skov et al., 2024) and, in Sub-Saharan Africa, in Ghanaian maize amended with fine-grained silicate rock flour (Oppong Danso et al., 2025; Akortey et al., 2026). It is this coupling of a public good and a private good in one practice that motivates analyzing the two returns together.

CDR potential and cost. A large literature estimates the carbon-removal potential and cost of ERW. Early techno-economic work derived global potentials and \$/t CO₂ cost curves and identified grinding energy and feedstock chemistry as the dominant cost and efficiency levers (Strefler et al., 2018), and continental assessments place croplands at the center of feasible deployment (Beerling et al., 2020, 2025a). Subsequent modeling has sharpened how climate governs weathering rates and hence the spatial distribution of potential (Baek et al., 2023; Deng et al., 2023). A parallel and still unsettled strand concerns how much of the gross weathering signal is durable: bicarbonate can re-precipitate as pedogenic carbonate or fail to export, so realized removal can fall well below stoichiometric ceilings (Renforth, 2019; Kanzaki et al., 2023), and recent reviews stress the resulting quantification and permanence uncertainties and the monitoring needed to resolve them (Schiedung et al., 2026; Levy et al., 2024; Beerling et al., 2025b). This body of work is rich on the geochemistry but typically frames ERW as a carbon supply curve, abstracting from the farm economy in which deployment must actually occur.

Application rates in comparable assessments. Spatially explicit assessments of ERW on croplands have almost universally applied a *uniform* basalt rate, and those rates cluster in a narrow band (Table 1). Global and regional models use 10 to 50 t/ha, often annually: 10 t/ha/yr in Baek et al. (2023) and Tu et al. (2026), 40 t/ha/yr in Beerling et al. (2020) and Beerling et al. (2025a), and 50 t/ha in the life-cycle and techno-economic assessment of Kroeger et al. (2026). Field trials test a wider range, from 10 to 20 t/ha in temperate (Skov et al., 2024) and West African (Oppong Danso et al., 2025) experiments to 50 to 100 t/ha

Table 1: Basalt application rates in comparable ERW assessments.

Study	Scope	Application rate
Baek et al. (2023)	Global croplands (model)	10 t/ha/yr
Tu et al. (2026)	Global adoption/equity (model)	10 t/ha
Beerling et al. (2020)	Global croplands (model)	40 t/ha/yr
Beerling et al. (2025a)	US croplands (model)	40 t/ha/yr
Kroeger et al. (2026)	Life-cycle / techno-economic	50 t/ha
Skov et al. (2024)	Temperate field trial (oat)	20 (up to 100) t/ha
Oppong Danso et al. (2025)	Ghana field trial (maize)	10, 50 t/ha
Akortey et al. (2026)	Ghana field trial (maize)	50, 100 t/ha
Jordan et al. (2026)	India on-farm (rice)	25–50 t/ha
This study	SSA, spatial (private + public)	targeted (lime req.) + uniform 10/20/50 t/ha

([Akortey et al., 2026](#)), and on-farm smallholder deployments in India apply 25 to 50 t/ha with documented yield and income gains ([Jordan et al., 2026](#)); reviews note experimental rates as high as several hundred t/ha ([Schiedung et al., 2026](#)). Our uniform scenarios (10/20/50 t/ha) therefore span the modeling mainstream, with our maximum at its upper edge and well below extreme experimental doses. Our targeted allocation departs from this literature: instead of a fixed uniform rate, it sizes the dose to each soil’s lime requirement, which is precisely what couples the carbon and agronomic accounts and motivates the net-export correction of Section 3.4.

Soil acidity and the economics of liming in SSA. Because ERW is, agronomically, “liming with a different rock,” the soil-acidity-remediation literature provides our closest methodological template. Acid soils are widespread across SSA and impose large, quantifiable yield penalties; recent continental work estimates that acidity depresses yields on 32.7 Mha, about 23% of cropland, a roughly US\$6 billion burden, a substantial share of which could be profitably alleviated by liming on the order of 6 to 9 Mha ([Vasco Silva et al., 2025](#)). That assessment builds on spatially explicit lime-requirement models ([Aramburu-Merlos et al., 2023](#)) and connects to farm-level analyses of the multi-year economics and

greenhouse-gas trade-offs of liming, for example in western Kenya ([Hijbeek et al., 2021](#)). We adapt this integrated, spatially explicit framework, reconfiguring its costs and crop-response functions for basalt and adding the carbon-removal benefit and the delivered-cost geography that ERW introduces.

Adoption and smallholder finance. Whether any of this potential is realized depends on who can finance it. The defining feature of the ERW investment, as with liming, is a large upfront outlay (on the order of \$600/ha for basalt) against benefits that accrue over several seasons. For credit-constrained smallholders with short decision horizons, such front-loaded investments are precisely those that the technology-adoption literature identifies as systematically under-adopted absent finance, insurance, or subsidy ([Jack, 2013](#)). This makes the locus of returns a first-order question: where farmers themselves capture enough value to pay, deployment can proceed through input markets or cost-sharing arrangements; where they do not, it requires external finance. Estimating the private return is therefore not only of academic interest but directly informs which financing instrument is feasible where.

Voluntary carbon markets and additionality. The most discussed external financing channel for ERW is the voluntary carbon market, where buyers increasingly favor removals that are durable, well quantified, and accompanied by development co-benefits. Two considerations bear directly on our analysis. First, integrity and measurement: because durable removal is uncertain and partly offset by soil processes, conservative net accounting matters for what can credibly be sold ([Schiedung et al., 2026](#); [Levy et al., 2024](#)). Second, additionality: a credit is meant to fund removal that would not otherwise occur. Coupled private and public returns complicate this test in an informative way. Where yield returns alone already justify the practice, carbon credits are weakly additional; where neither return alone suffices but their sum does, carbon finance is pivotal and clearly additional. The spatial structure of the two returns is thus not incidental to crediting but central to it.

What is missing. Despite rich literatures on ERW geochemistry, on the economics of liming, and on carbon-market design, no prior work, to our knowledge, estimates the spatial distributions of the private (yield) and public (carbon) returns to ERW side by side, characterizes where each is sufficient on its own and where only their combination is, or frames the carbon credit as the marginal financing that converts an otherwise-unprofitable yield-improving practice into a viable one, particularly in a high-acidity, smallholder-dominated setting like SSA. Nor has the durable-removal accounting that this coupling demands, in which alkalinity consumed neutralizing soil acidity is not credited as exported carbon, been applied spatially. We address these gaps.

3 Conceptual framework and modeling approach

3.1 Private and public returns

Consider a representative cropland pixel and a one-off decision to apply crushed basalt at an agronomically calibrated rate. The per-hectare gross margin of the practice has three economic components: an agronomic return from the yield response to acidity neutralization, R_a ; a carbon return from selling the verified CO₂ removed, R_c ; and the delivered cost of the basalt, C (quarrying, grinding, transport, and spreading). All three are spatially heterogeneous. We write the combined gross margin as

$$\text{GM} = R_a + R_c - C, \quad R_c = \rho(p - m), \quad (1)$$

where ρ is the net CO₂ removed per hectare (t CO₂/ha, after life-cycle emissions, pedogenic-carbonate loss, and the alkalinity consumed neutralizing soil acidity; Section 3.4), p is the carbon price (\$/t CO₂), and m is the per-tonne MRV cost.

The novelty is not the combined margin but its *decomposition* into a private and a public

test, each charged the *full* cost of the practice:

$$\text{Private-sufficient: } \text{GM}_a \equiv R_a - C > 0, \quad (2)$$

$$\text{Public-sufficient: } \text{GM}_c \equiv R_c - C > 0. \quad (3)$$

Equation (2) identifies land where a farmer would rationally invest on yield grounds alone, ignoring carbon; equation (3) identifies land where a carbon buyer would fund the practice on removal grounds alone, ignoring yield. Their *intersection* ($\text{GM}_a > 0 \wedge \text{GM}_c > 0$) is land where the investment is doubly justified – and, crucially, where the carbon credit is *least additional*, because the agronomic case already stands on its own. This yields a five-way typology of cropland (Figure 1): both, private-only, public-only, *combined-only* (neither stream alone covers C but together they do, $\text{GM} > 0$), and neither. The combined-only set is where carbon finance is pivotal: it is additional by construction and it is precisely the land that VCM revenue moves from unprofitable to profitable.

The public envelope is a marginal-abatement-cost test. Substituting $R_c = \rho(p - m)$ into (3) and noting that the per-hectare application rate cancels between ρ and C (both scale linearly with tonnes of rock applied), public-sufficiency reduces to

$$\underbrace{\frac{c_t}{\delta_t}}_{\text{MAC}} < p - m, \quad (4)$$

where c_t is the delivered cost per tonne of basalt and δ_t is the CO_2 removed per tonne of basalt. The public envelope is therefore exactly the land whose marginal abatement cost (MAC) lies below the net carbon price. Because δ_t – the realized removal per tonne of rock – is small and uncertain, equation (4) makes clear that the carbon case is governed first by the *CDR rate*, and only second by the credit price. This is the analytical core of the paper’s main empirical result (Section 4.3).

3.2 Data and raster modeling framework

We build on a per-pixel ($\sim 0.083^\circ$, ~ 9 km) *ex ante* model covering 44 SSA countries and 23 SPAM crops. Soil properties (pH, exchangeable acidity and bases, bulk density) come from SoilGrids (Poggio et al., 2021); harvested area, production, and yield from SPAM (Yu et al., 2020); climate from WorldClim (Fick and Hijmans, 2017); basalt source lithology from GLiM (Hartmann and Moosdorf, 2012); a motorized friction surface for haulage cost-distance from Weiss et al. (2018); and producer prices from FAOSTAT (FAO, 2023) (2016–2020 medians). Lime-equivalent requirements use the LiTAS method (Aramburu-Merlos et al., 2023); the agronomic suitability response uses EcoCrop. Full layer provenance and constants are documented in the model repository.

3.3 Cost, carbon, and agronomic terms

Delivered cost per tonne of basalt is the sum of a quarry-gate price (\$10/t), grinding (energy use $\times$ country electricity tariff), spatial transport (travel-time $\times$ haulage rate), and spreading (\$8/t). Net CO₂ removal per tonne combines feedstock stoichiometry, a Lewis-calibrated reactive fraction, a grain-size factor, a climate (temperature–moisture–pH) scalar, and a deduction for alkalinity that re-precipitates locally as pedogenic carbonate; it is then netted of grinding, transport, and spreading life-cycle emissions. The agronomic return is the modeled yield response to acidity neutralization valued at crop-specific producer prices. We adopt baseline parameters of a \$150/t CO₂ credit price, \$20/t CO₂ MRV cost, 50 μ m grain size, and a 10% discount rate. Table 2 decomposes the delivered cost.

3.4 Net-export carbon accounting

The alkalinity released by weathering can either neutralize soil acidity or export as bicarbonate to the ocean; it cannot do both. Alkalinity that neutralizes exchangeable acidity re-releases the captured CO₂ ($\text{HCO}_3^- + \text{H}^+ \rightarrow \text{H}_2\text{O} + \text{CO}_2$, with protons supplied by Al^{3+}

Table 2: Delivered-cost decomposition per tonne of basalt, with the separate MRV deduction. Delivered cost = quarry-gate + grinding + transport + spreading; MRV is netted from the carbon price rather than added to delivered cost. Transport is the only spatially varying (variable) term and drives the cost heterogeneity that determines where ERW is deployable; grinding cost varies with the country electricity tariff. Life-cycle CO₂ of grinding, transport, and spreading is separately deducted from gross removal (not shown).

Component	Reference value	Type	Basis / source
Quarry-gate feed-stock	\$10/t	fixed	basalt fines as aggregate by-product
Grinding (50 μ m)	19.4 kWh/t $\times$ tariff ($\approx$ \$1.9/t)	fixed	Strefler et al. (2018) energy curve
Transport (haulage)	travel-time $\times$ \$0.04/min/t	variable	30 t truck over MAP friction surface
Spreading	\$8/t	fixed	on-farm application
Delivered cost/t	quarry + grind + transport + spread		consumed per pixel
MRV (netted separately)	\$20/t CO ₂	per t CO ₂	Levy et al. (2024) pilot benchmark \$30

hydrolysis), exactly as agricultural lime does – which is why lime is not a carbon-removal technology. Durable removal is therefore only the alkalinity that exports *beyond* the soil’s acidity demand. We credit

$$\rho_{\text{net}} = \max(\rho_{\text{gross}} - F S, 0), \quad (5)$$

where S is the soil’s exchangeable acidity expressed as a lime requirement (t CaCO₃/ha, from the LiTAS/Kamprath calculation already in the pipeline) and $F = 0.88$ t CO₂ per t CaCO₃-eq is the CO₂ re-released per unit of acidity neutralized – fixed by the feedstock’s own stoichiometry, not tuned. The sink is regime-specific: the *standing* exchangeable acidity at first application, but only the annual *re-acidification* in steady state. Equation (5) is a first-order (sequential) bound: it assumes the acidity sink is filled before any alkalinity exports, whereas in reality weathering, neutralization and leaching occur together, so the truth lies between ρ_{net} and ρ_{gross} ; we report both. A direct corollary, central to the results, is that the agronomically optimal (lime-requirement) dose is CDR-minimal: durable removal requires applying *beyond* the acidity demand, so the private and public objectives pull the

application rate in opposite directions.

3.5 Over-liming penalty

The same surplus alkalinity that exports as durable carbon also raises soil pH. Where a dose pushes pH past a crop’s optimum, yield declines through micronutrient lockout and phosphorus retrogradation. Our baseline credits the yield response to acidity relief but does not penalize this overshoot; as a refinement we evaluate each crop’s pH suitability at the *post-application* pH. We estimate that pH from the reacted alkalinity (equal to ρ_{gross} on the same F basis), which first neutralizes the soil’s exchangeable acidity and, once that demand is met, raises pH into the alkaline range buffered by the soil’s ECEC; a crop-specific EcoCrop pH ceiling then haircuts the agronomic return above the crop’s optimum. This is a first-order refinement whose magnitude depends on the soil buffer constant and the realized reactive fraction, which we carry as a sensitivity band (Section 4.2).

3.6 Temporal regimes and allocation rules

We report three temporal accountings: *year-1* (undiscounted, agronomic return summed over the 5-year basalt residency), *NPV* at 10% (carbon revenue phased over five years and discounted; agronomic NPV over the reapplication interval), and *equilibrium* (steady-state maintenance dose). Because durable carbon removal is intrinsically a steady-state quantity (Section 3.4), we lead both the public and the private case on the *equilibrium* regime – which places the two return streams on the same recurring, per-period footing – and report year-1 and NPV as robustness. For allocation we contrast a spatially *targeted* LiTAS dose (applied only to acid pixels above each crop’s tolerance) with a *uniform* 20 t/ha dose across all cropland, the standard convention in the ERW modeling literature.

3.7 Envelopes, aggregation, and robustness

For each pixel we compute the area-weighted (across co-located crops) agronomic return, basalt cost, and carbon revenue, and evaluate (2)–(3). We summarize each envelope by harvested area and by the aggregate private return (value of ERW-induced extra output), public return (durable carbon revenue), and delivered cost summed over its pixels; we also report an approximate farm count (affected hectares divided by national average agricultural holding size; Lowder et al., 2016) in the country-level tables. Throughout, area shares are taken relative to the *treated* cropland—the 17.6 Mha of acid-eligible land on which the targeted dose is applied, our action set—rather than all cropland.¹ Because the per-hectare dose cancels, the public envelope, the MAC surface ((4)), and the entire parameter sweep are computed analytically from three per-pixel primitives, with no model re-run. We probe robustness over delivered cost ($\times 0.5$ – 2.0), carbon price (\$50–250), yield benefit ($\times 0.5$ – 2.0), and CDR rate ($\times 0.5$ – 1.5), and define a pixel’s robustness score as the fraction of that grid under which it remains in the intersection. All analysis is reproducible via the `erw-11` script in the repository.

4 Results

4.1 The private, public, and intersection envelopes

Table 3 reports the three envelopes plus the combined margin in the equilibrium steady state with net-export carbon accounting. Under the targeted allocation the private (yield) envelope is the largest at 4.3 Mha, the durable public (carbon) envelope is 1.7 Mha, and the two coincide on 1.2 Mha. The combined envelope is larger still at 6.9 Mha. Decomposing targeted cropland into the typology of Section 3.1 (Figure 1): roughly 3.1 Mha is private-only, 0.5 Mha public-only, 1.2 Mha both, and **2.1 Mha is combined-only** – unprofitable

¹Total SSA harvested cropland is 146 Mha (SPAM). Shares of that broader base are roughly one-eighth of the treated-cropland shares reported here (17.6/146).

on yield alone and on carbon alone, but profitable when the two revenue streams are stacked. The intersection is where a carbon purchase is *strengthened* by a co-benefit but the credit is least additional; the combined-only band is where carbon finance is *pivotal* – additional by construction, and precisely the land that VCM revenue moves from unprofitable to profitable. This is the paper’s central economic result: across most ERW-viable land in SSA, voluntary carbon finance is not a windfall on an already-profitable practice but the marginal revenue that makes an otherwise unprofitable agronomic investment pencil out.

The allocation comparison makes the agronomy–carbon dose trade-off concrete (Table 3). Moving from the targeted (lime-requirement) dose up the uniform ladder, the private envelope shrinks monotonically (4.3, 3.5, 1.3 and 0.5 Mha at targeted, 10, 20 and 50 t/ha), because a fixed dose increasingly wastes rock on low-response soils, while the durable public envelope and especially its carbon yield grow (public area 1.7, 2.8, 3.2 and 3.4 Mha; durable CDR 4.8, 6.2, 15 and 42 Mt), because applying beyond the acidity demand leaves more alkalinity free to export. The combined footprint peaks near the agronomic dose (6.9–7.1 Mha at targeted and uniform-10) and the carbon yield at uniform-50, the latter at far higher cost. This is the dose-design corollary of Section 3.4: maximizing private return points to the lime requirement, maximizing durable removal points to over-application, and the two coincide only where the surplus pays for itself. We therefore report the full uniform ladder alongside the targeted dose, which we retain as the agronomically motivated default.

4.2 Durable carbon and the agronomy–carbon trade-off

The carbon figures above already net out the alkalinity consumed neutralizing soil acidity (Section 3.4), and that deduction is first-order, not cosmetic. At first application the standing acidity absorbs most of the weathered alkalinity, so only about one-fifth of gross removal is durably exported (targeted: 81 Mt gross $\rightarrow$ 19 Mt net; uniform: 101 $\rightarrow$ 19 Mt). In the equilibrium steady state, where the maintenance dose offsets only annual re-acidification, most removal survives (targeted: 26 $\rightarrow$ 19 Mt, 73%; uniform: 101 $\rightarrow$ 88 Mt, 87%). Durable

Table 3: Private, public, and combined return envelopes in the equilibrium steady state with net-export carbon accounting, by allocation rule (targeted; uniform 10/20/50 t/ha). *Area* is harvested hectares (Mha). *Private* is the value of additional crop output from ERW; *Public* is durable net-export CDR valued at \$150/t CO₂ net of the \$20/t CO₂ MRV cost; *Cost* is delivered basalt (quarry, grinding, transport, spreading). Monetary totals are \$M per maintenance cycle, summed over the pixels in each envelope; combined gross margin = Private + Public – Cost. *CDR* is durable net-export removal (Mt).

Allocation	Envelope	Area	Private (\$M)	Public (\$M)	Cost (\$M)	CDR (Mt)
Targeted	Private	4.31	3,496	929	1,743	7.1
Targeted	Public	1.65	1,339	630	434	4.8
Targeted	Intersection	1.20	1,274	510	337	3.9
Targeted	Combined > 0	6.85	4,311	1,418	2,788	10.9
Uniform 10	Private	3.52	3,277	710	1,432	5.5
Uniform 10	Public	2.75	1,364	809	610	6.2
Uniform 10	Intersection	1.15	1,244	371	258	2.9
Uniform 10	Combined > 0	7.09	4,018	1,567	2,667	12.1
Uniform 20	Private	1.29	1,876	702	849	5.4
Uniform 20	Public	3.16	1,422	1,965	1,424	15.1
Uniform 20	Intersection	0.74	1,124	520	330	4.0
Uniform 20	Combined > 0	5.13	2,973	2,758	3,165	21.2
Uniform 50	Private	0.45	997	796	519	6.1
Uniform 50	Public	3.38	1,452	5,438	3,850	41.8
Uniform 50	Intersection	0.41	890	754	439	5.8
Uniform 50	Combined > 0	3.84	1,827	6,020	4,632	46.3

removal from ERW on acid soils is thus largely a *steady-state* phenomenon; the first application is, to first order, liming with a small carbon tail – which is why we evaluate the public case at equilibrium.

The same accounting resolves how returns vary with soil pH. Both the yield benefit and the gross weathering rate are largest on the most acidic soils, so a naive model would place *both* returns on the same low-pH land. Once the acidity sink is deducted, the durable-carbon return shifts toward *less* acidic soils – where less alkalinity is spent on neutralization and more exports – so the private and public returns become spatially complementary (Figure 3). The corollary for deployment design: because the lime-requirement dose is sized to the acidity demand, it is carbon-minimal; maximizing durable removal requires applying *beyond* agronomic need, so the two objectives imply different optimal rates and the carbon payment must cover the surplus rock that yields no extra harvest.

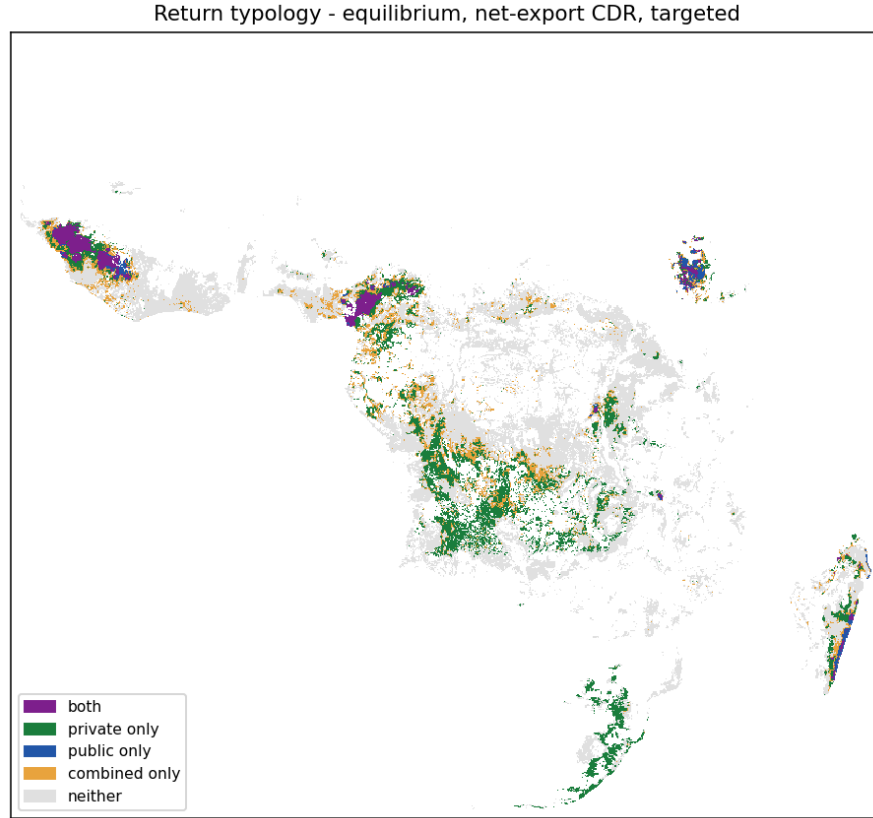


Figure 1: Return typology of SSA cropland in the equilibrium steady state with net-export carbon accounting, targeted allocation. Purple = both private and public returns independently sufficient (intersection); green = private-only; blue = public-only; orange = combined-only (viable only when both streams are stacked); grey = neither.

Over-application also carries an agronomic cost that the base model omits: pushing soil pH past the crop optimum reduces yield (Section 3.5). Charging this over-liming penalty leaves the private return essentially unchanged at the targeted dose but erodes it at high uniform rates, and it does so with sharply increasing uncertainty (Figure 4). Centrally, the penalty reduces the aggregate private return by about 7% at the targeted dose, 6% at uniform-10, 5% at uniform-20, and 18% at uniform-50; but the plausible range widens from a tight 5–8% at the agronomic dose to fully 3–99% at uniform-50, depending on soil buffering and realized reactivity. The direction is robust while the high-rate magnitude is not: blanket high rates trade a bounded carbon gain for a large and poorly-bounded yield risk, whereas the targeted dose avoids the exposure. This is the third margin – alongside

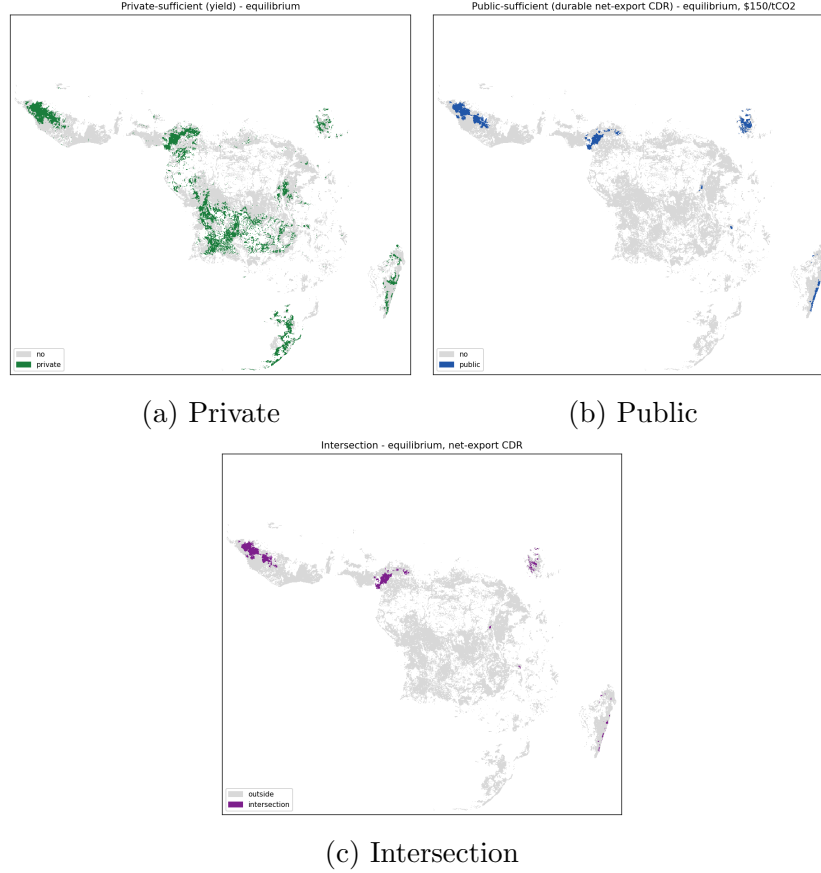
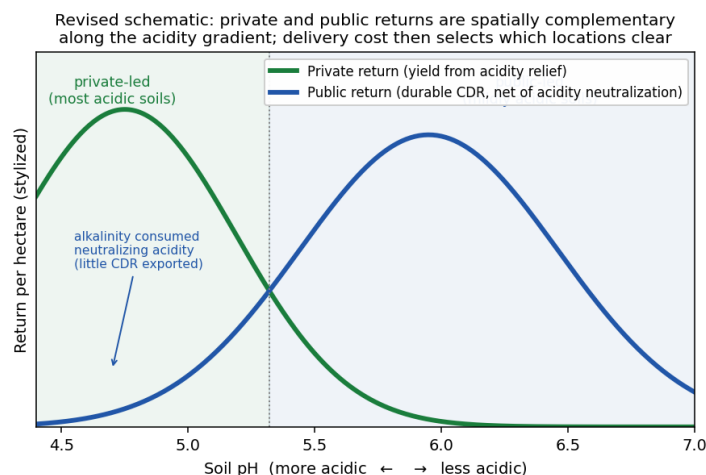


Figure 2: The three envelopes mapped separately (equilibrium steady state, net-export carbon accounting, targeted). Private: agronomic return exceeds delivered cost. Public: durable net-export carbon revenue at \$150/t CO₂ exceeds delivered cost. Intersection: both hold.

rising cost and the acidity-sink deduction on CDR – on which over-application is disfavored, and it completes the case for targeting plus modest over-application rather than uniformly heavy rates.

4.3 What binds the public (carbon) envelope

Because ERW investment interest concentrates on the carbon case, we diagnose it directly through the MAC test of equation (4), on durable net-export removal. In the equilibrium steady state the area-weighted MAC has a median of \$446/t CO₂ (Figure 5a), against a net credit price of only \$130 (\$150 less \$20 MRV); only the cheapest fraction of land clears, which



Realized returns are additionally gated by the delivery-cost gradient (distance to basalt): within each band, only low-cost, basalt-proximate pixels clear.

Figure 3: Private and public returns are spatially complementary along the soil-pH gradient once durable (net-export) carbon is credited: private (yield) returns peak on the most acidic soils; public returns peak on mildly acidic soils where weathered alkalinity exports rather than neutralizing acidity. Realized returns are then further selected by the delivery-cost gradient.

is why the durable public envelope is 1.7 Mha. The contrast with first application is stark: under net-export accounting the year-1/NPV median MAC rises to roughly \$1,250/t CO₂ and the carbon case all but disappears, because the standing acidity must be paid down before any carbon exports – again locating durable removal in the steady state.

The durable public envelope of 1.7 Mha is 9.4% of the treated cropland (1.1% of all SSA cropland), and its size is governed by the carbon price rather than by any kinetic assumption: carbon-alone deployable area rises steeply as the price climbs, from 2% of treated cropland at \$100/t CO₂ to 9% at today's \$150, 28% at \$350, and 50% at \$500 (Table 4). Two implications follow. First, the physical removal potential is large but its *economic* viability under carbon revenue alone is price-limited: at present prices durable CDR pays for its delivered cost on only about a tenth of the ERW-relevant cropland. Second—and this is the crux of the argument—the fact that carbon revenue rarely clears on its own is precisely why the private co-benefit matters. Land that carbon alone cannot reach at \$150 becomes viable once the yield return is stacked on top (the combined-only band, Section 4.1); the co-benefit is not a nicety but the mechanism that mobilizes voluntary carbon finance across the region.

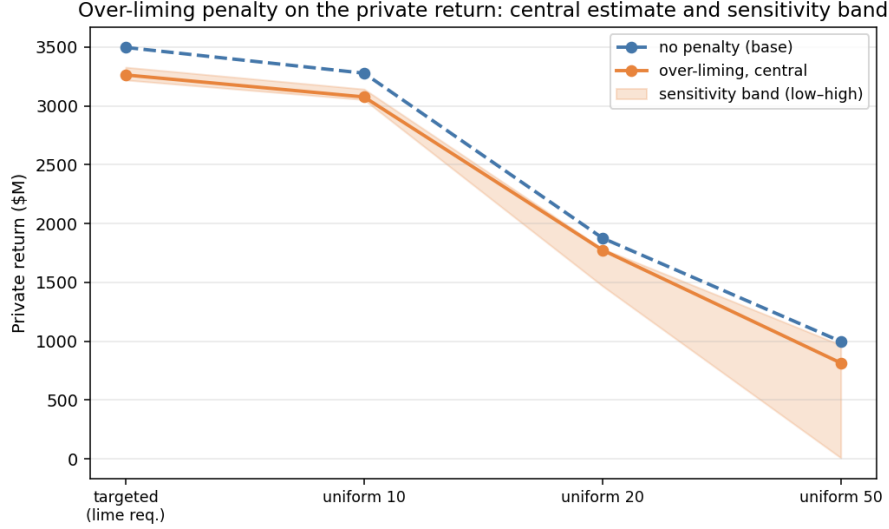


Figure 4: Over-liming penalty on the aggregate private return across the application-rate ladder: no-penalty base, central estimate, and the sensitivity band spanning soil-buffer and realized-reactivity assumptions. The band is tight at the agronomic (targeted) dose and fans out at high uniform rates.

And because the marginal abatement cost is the ratio of delivered cost to CO₂ removed per tonne, the two routes to broadening carbon-alone viability are a higher realized CDR rate and cheaper delivery, both of which lower it.

The binding factor is the realized CO₂ removed per tonne of rock, δ_t (here on the order of 0.1 to 0.25 tonnes CO₂ per tonne of rock), not the credit price or logistics. Figure 6 makes this quantitative (shown on the gross NPV basis for comparability with the parameter grid): doubling δ_t raises the public envelope far more than a 33% higher carbon price or a 33% delivered-cost reduction. δ_t is set by the reactive fraction, grain size, the climate factor, and pedogenic-carbonate loss – the parameters that current process knowledge and MRV resolve least well. The economic and the scientific frontier therefore coincide: the largest lever on the carbon case is also the least-resolved quantity.

4.4 Cost-side levers: at-scale solar haulage

Transport is the largest single component of delivered cost (~56% on average, median ~65%) and is wholly variable in distance. We evaluate an at-scale solar-electric haulage scenario that

Table 4: Carbon-alone deployability is price-conditional. Public-sufficient area (durable net-export CDR revenue exceeds delivered cost) in the equilibrium steady state, targeted allocation, across VCM carbon prices. Shares are of the 17.6 Mha treated cropland; the all-cropland share (of 146 Mha) is shown for reference.

VCM price (\$/t CO ₂)	Public area (Mha)	% treated	% all cropland	Durable CDR (Mt)
100	0.35	2.0	0.2	1.7
150	1.65	9.4	1.1	4.8
250	2.51	14.3	1.7	6.1
350	4.84	27.6	3.3	11.0
500	8.76	49.9	6.0	15.8

displaces the fuel/energy portion of trucking cost. Grounding the energy share in trucking-cost data and the diesel-to-solar-electric energy gap implies a feasible $\sim 25\%$ reduction in the transport term (range 15–40%), i.e. $\sim 14\%$ off average delivered cost. Applied to the grid (NPV basis, for comparability with the sensitivity grid), this widens the *private* envelope materially (2.0 to 2.6 Mha at -25% , to 3.1 Mha at -40%) but the public envelope only slightly (1.8 to 1.9 Mha), confirming the asymmetry implied by equation (4): cost reductions relax the private test directly but relax the public test only to the extent the MAC falls below the credit price. Carbon price and CDR rate move the public envelope; logistics cost moves the private one. The two policy levers are thus complementary and target different margins.

4.5 Targeting and robustness

Repeating the envelope construction across the full parameter grid yields a per-pixel robustness score (Figure 7). Under the NPV regime no pixel survives the entire deliberately wide grid; the intersection is inherently parameter-sensitive. A core-targeting tier (in the intersection under at least half the grid) collapses to a small, high-confidence set concentrated in the Cameroon highlands, while a broader candidate tier (at least one-third of the grid) covers ~ 0.8 Mha, ~ 0.5 M farms, and $\sim \$0.75$ B of crop value, extending into the Guinean and Ethiopian highlands – areas where acidic soils, a favorable weathering climate, basalt proximity, and adequate crop prices coincide.

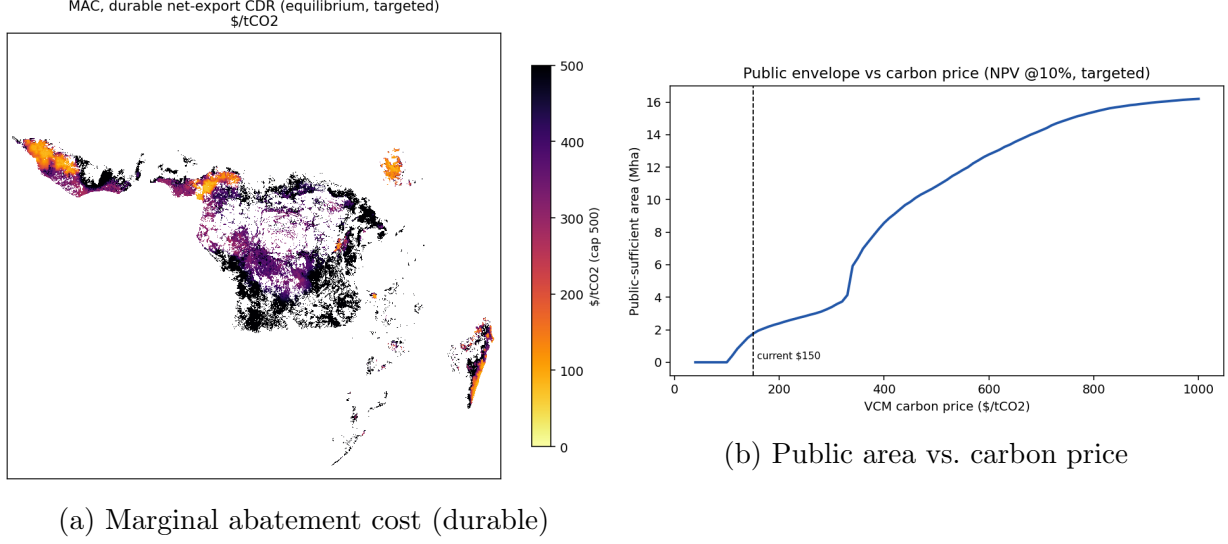


Figure 5: (a) Marginal abatement cost of durable net-export ERW-CDR ($\$/t\ CO_2$, capped at 500; equilibrium, targeted). (b) Public-sufficient area as a function of the VCM carbon price (gross NPV accounting, an upper bound); the schedule is steeply convex around current prices.

Table 5: Public envelope and total credited CDR (targeted), under gross versus net-export accounting, at first application (NPV) and in steady state (equilibrium). Net-export accounting collapses the first-application carbon case and locates durable removal in the steady state.

Accounting	Public, NPV (Mha)	Public, equil. (Mha)	CDR, NPV (Mt)	CDR, equil. (Mt)
Gross	1.75	2.23	81	26
Net-export	0.14	1.65	19	19

The carbon case behaves very differently across regimes once durable removal is credited (Table 5). Under gross accounting the public envelope looks broadly similar at first application and in steady state; under net-export accounting it nearly vanishes at first application (NPV public $1.75 \rightarrow 0.14$ Mha), because the standing acidity consumes the alkalinity, and is restored only in the equilibrium steady state ($2.23 \rightarrow 1.65$ Mha), where the maintenance dose offsets only annual re-acidification. The private envelope, which does not depend on the carbon accounting, is stable across regimes. This regime-dependence is precisely why we lead the carbon case on equilibrium.

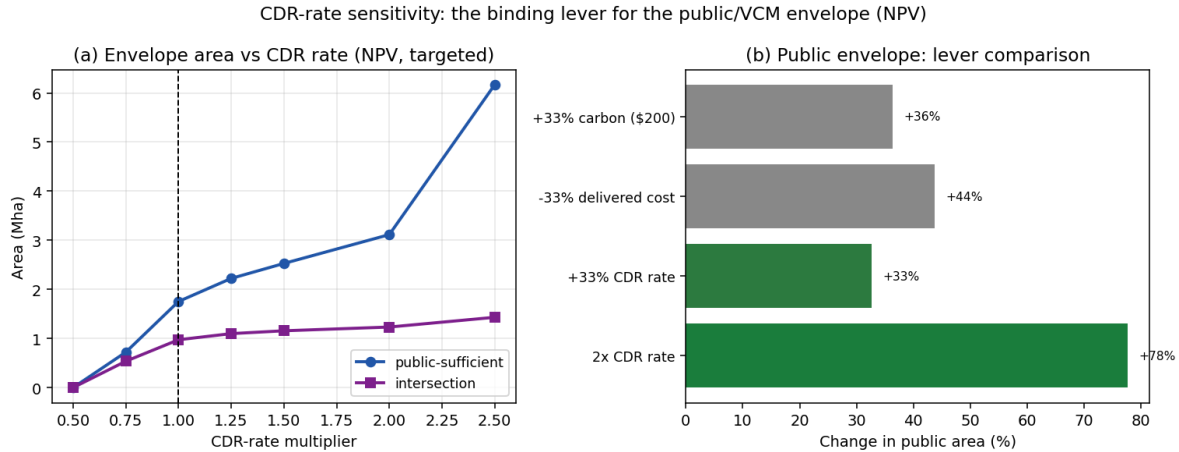


Figure 6: The CDR rate is the binding lever for the public envelope. (a) Public and intersection area versus the realized CDR-rate multiplier. (b) Change in public-sufficient area under alternative levers: doubling the CDR rate dominates plausible carbon-price or delivered-cost moves.

5 Discussion

Three implications follow for the design of ERW programs and for the agricultural economics of carbon-coupled practices more generally.

Carbon finance is the marginal, not the windfall, revenue. The dominance of the combined-only band means that on most ERW-viable SSA cropland the agronomic return and the carbon return are each individually insufficient to cover the upfront cost of crushed basalt, but their sum is not. Voluntary carbon revenue is therefore best understood as the financing that crosses the adoption threshold for a yield-enhancing practice that farmers would otherwise forgo – a framing closer to a cost-share or input subsidy than to a pure carbon transaction. This both strengthens the additionality claim on that land (the practice does not happen without the payment) and reframes the distributional question: the credit buyer effectively subsidizes a private yield gain, and program design must decide how that joint surplus is split between buyer and farmer.

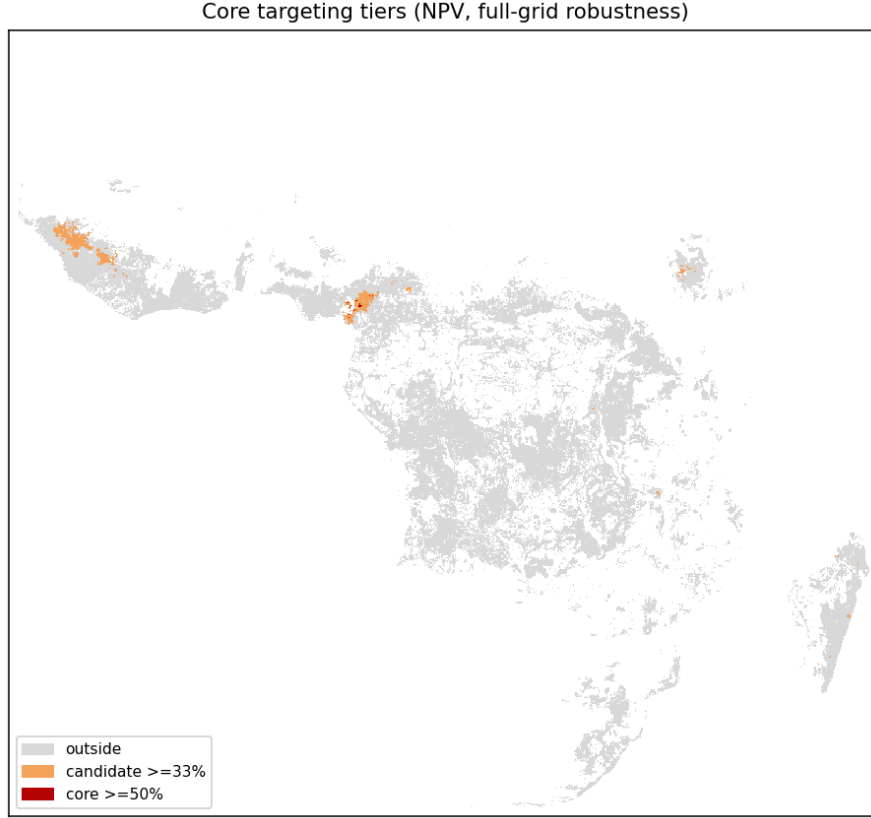


Figure 7: Core-targeting tiers under the NPV regime. Red = robust core (intersection under $\geq 50\%$ of the parameter grid); orange = candidate tier ($\geq 33\%$). The core concentrates where soil acidity, weathering climate, basalt access, and crop value coincide.

The carbon case is CDR-quantification risk, not price risk. Equation (4) and Figure 6 show that the public envelope hinges on the realized removal per tonne of rock, the single largest lever and the least-resolved scientific quantity. For investors this relocates the central risk from the carbon market (price) to measurement, reporting, and verification (quantity and durability). It also reframes MRV: the \$20/t CO₂ MRV cost is not merely a deduction but the mechanism that converts an uncertain δ_t into a sellable, high-integrity tonne. Investment in cheaper, tighter measurement raises the *effective* CDR rate that can be monetized – exactly where the leverage lies.

Targeting and sequencing. The robustness analysis points to a small core geography that is attractive across a wide range of assumptions, and a broader candidate tier for early

deployment. Because cost-side and revenue-side levers move different envelopes, an efficient program pairs logistics investment (basalt siting, low-carbon haulage) where the private case is close, with carbon-price or CDR-rate improvements where the public case is close.

Limitations. The year-1 accounting is the most optimistic and the NPV the most conservative; we report both. Farm counts use national average holding sizes and are order-of-magnitude. The agronomic response and the climate and pedogenic-carbonate terms remain empirical proxies; the CDR-rate sweep is our stand-in for that structural uncertainty, and tighter field calibration is the clear priority. We do not model tenure, who finances the upfront cost, compaction or N₂O effects, or country-level credit eligibility (Article 6), each of which would refine the deployable geography.

6 Conclusion

Enhanced Rock Weathering is unusual among carbon-removal options in paying a farmer twice: once in yield and once in carbon. Mapping those two returns separately across Sub-Saharan Africa shows that they rarely each suffice on their own – but that together they make a meaningful area of cropland profitable that neither would alone. The policy object of interest is therefore the *combined-only* land, where voluntary carbon finance is the marginal revenue that unlocks a yield-improving investment, and the binding technical constraint is how much CO₂ a tonne of rock verifiably removes. Treating ERW as a coupled agronomic-and-carbon proposition, rather than as a pure CDR supply curve, changes both where one would deploy it first and what one would invest in to expand it.

Data and code availability

All input layers are public. The analysis pipeline, the envelope and robustness scripts (`erw-1-erw-11`), and the code that reproduces the figures and tables reported here are avail-

able in the project repository `erw-exante-v2` (<https://github.com/chamb244/erw-exante-v2>). Large input rasters are distributed separately on request; every raster’s public source is documented in the repository.

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